

DEWS: A Live Visual Surveillance System for Early Drowning Detection at Pool

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Abstract—A real-time vision system operating at an outdoor swimming pool is presented in this paper. The system is designed to automatically recognize different swimming activities and to detect occurrence of early drowning incidents. We have named this system the *Drowning Early Warning System (DEWS)*. One key challenge we faced in the problem is the relatively high level of noise in the steps of foreground detection and behavior recognition. Therefore, a set of methods in the fields of background subtraction, denoising, data fusion and blob splitting are proposed, which have been motivated by characteristics of aquatic background and crowded scenario at the pool. In the step to detect an early drowning incident, visual indicators of distress and drowning are incorporated through a set of foreground descriptors. A module comprising data fusion and hidden Markov modeling is developed to learn unique traits of different swimming behaviors, in particular, those early drowning events. The experiment of this work reports realistic on-site evaluations performed. Examples of interesting behaviors, i.e., distress, drowning, treading and numerous swimming styles, are simulated and collected. Experimental results show that we have established a prototype system which is robust and beyond the stage of proof-of-concept.

Index Terms—Crowded scenario, dynamic aquatic background, foreground silhouette extraction, modeling of water crisis behaviors, real-time video surveillance system.

I. INTRODUCTION

CURRENT development towards better security and safety has led to wide installation of closed circuit television (CCTV) systems, in both public and private places [1], [2]. With recorded footages and information being made available in large quantity, this has meantime raised the demand for an *automated* solution. Visual surveillance research aims to achieve this goal to provide real-time video content analysis and understanding capabilities.

A visual surveillance system typically involves complex tasks such as detection of targets under varying ambient conditions, si-

multaneous detection and tracking of multiple targets, and recognition of complex behaviors which vary each time they are performed [1]–[3]. These challenging tasks have traditionally led to research based on captured footages [4] or simulations done in a laboratory environment [5], [6] to have fundamental understanding of different vision components. While varied degrees of success are reported in these early studies, directly deploying these methods to real-world problems remains insufficient, particularly, the robustness to address long-hour operation and the technical complexity involved. Hence, this observation has motivated several recent works looking into live systems in *different real-world environments* and their *societal importance* [7]–[13].

In this paper, we address the problem of developing a *live* surveillance system at *pool* environment. It involves a new branch of visual surveillance study which is in an *aquatic* environment. The objectives here are to recognize different swimming activities and to provide constant monitoring to watch out for occurrence of any early drowning incidents. Thus, we have named the proposed system the Drowning Early Warning System (DEWS). To the best of our understanding, the proposed DEWS system is the first automated vision system using *overhead* cameras for automated pool monitoring [14]. The intended application is to provide an additional level of safety on top of the watchful eyes of lifeguards.

In our problem, water disturbance of aquatic background that exhibits random and continual movement poses new technical challenges. This has led to difficulties to distinguish between foreground or background movements and to recognize different swimming behaviors. In this paper, we focus the discussion on two basic aspects: *foreground silhouette extraction* and *behavioral recognition*. We begin this work by describing a statistical method that models background in terms of homogeneous blob movements. To detect different swimming activities, a framework for recognizing each activity trait is developed. The aim here is to recognize different types of swimming behaviors, in particular, early drowning events in a real environment setup.

In addition, we explore in this paper solutions to address: removal of highlights for operation at nighttime and separating contiguous blob of a few foreground targets into separate entities in order to handle moderate level of crowd.

This paper is organized as follows. Section II presents a detailed study of our problem and a review of relevant works. Section III describes a proposed foreground silhouette extraction module. In Section IV, details of a proposed water crisis inference scheme are explained. Section V describes a blob splitting mechanism to handle moderate level of crowd. Finally, in Sections VI and VII, experimental results and concluding remarks are reported.

Manuscript received October 20, 2006; revised February 8, 2007. This work was supported by the Enterprise Challenge Unit, Singapore's Prime Minister's Office and the Singapore Sports Council. This paper was recommended by Associate Editor D. S. Turaga.

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Digital Object Identifier 10.1109/TCSVT.2007.913960



Fig. 1. Proposed DEWS system installed at an outdoor public swimming pool.

II. OUR WORK IN PERSPECTIVE

A. Proposed DEWS System

Building a vision-based surveillance system for pool environment has attracted some attentions recently. Relevant works can be referred to [15]–[18], which proposed a system that utilizes *underwater* cameras to detect drowning event at its *later* stage, where the victim has sunk to the bottom of pool. The primary concept of this system is to detect motionless drowned victims.

From the point to yield a more stringent safety, detecting drowning event at its *early* stage while the victim is struggling at water surface sounds an attractive alternative. This motivates the exploration of using *overhead* cameras as shown in Fig. 1. The proposed system consists of overhead cameras mounted surrounding the pool. In the setup, each camera monitors a local region and there is overlapping between the views of two adjacent cameras. The combined view of all cameras then provides a full coverage of the pool.

The use of *overhead cameras* as proposed may do away and complement a few practical concerns that a system of using underwater cameras may have. For the setup of using underwater cameras, blocking of camera views by standing swimmers will become an issue when it is deployed at a shallow pool. This issue is of particular concern in our consideration since shallow pools are more commonly found in our region. Secondly, the use of underwater cameras may incur additional requirement of water proof feature, which may lead to higher installation and maintenance cost. On the other hand, the ability to monitor different swimming activities and detection of early drowning incidents based on overhead views can potentially contribute to a faster drowning detection.

B. Technical Challenges and Reviews

Before proceeding further, we firstly analyze the technical challenges of our problem and its key differences compared with other state-of-the-art problems. This will put our problem in perspective.

Fig. 2 presents a study of background movements at four arbitrary selected points on the image by analyzing the intensity distributions of these pixels. The horizontal axis of subfigures on the right represents gray level of pixel. Each row in the vertical axis represents intensity histogram of a selected pixel's values over 200 frames. Thus, the figure shows intensity distributions at the four selected points over 40 000 frames. When comparing with the problem of detecting human on the ground [8], wider

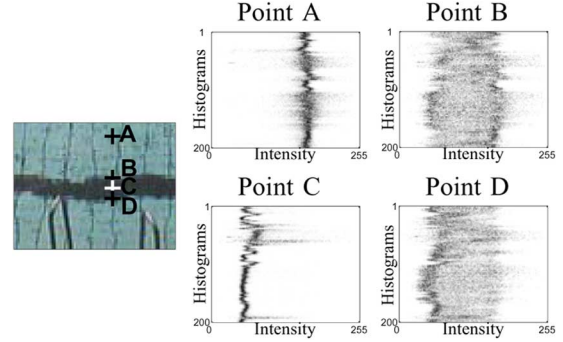


Fig. 2. Study of background movements at four arbitrary selected points on the image, labeled as Points A to D given on the left image. Widely distributed intensity histograms are observed for these four points due to continual and random movement of the aquatic background.

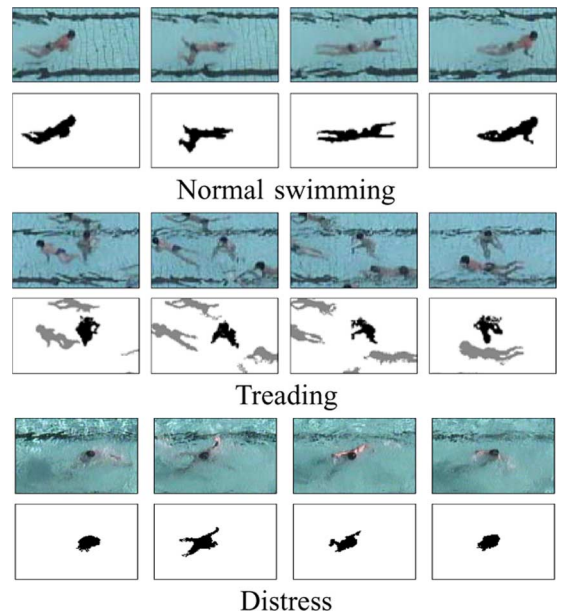


Fig. 3. Segmented silhouettes of foreground targets demonstrate: normal swimming, treading and distress. Foreground targets are marked in black color. (Odd rows show frames captured. Even rows show silhouette maps of segmented foreground.)

distributions are observed in our problem. This is due to continual and random movement of aquatic background which will contribute to greater level of noise to the steps of foreground subtraction and features extraction.

For behavioral recognition, one major challenge faced is the need to identify different complex behaviors from segmented targets which are typically of moderate size. This is due to the constraint where only a limited number of cameras can possibly be installed. Fig. 3 shows silhouettes of segmented targets demonstrating different types of swimming behaviors. Foreground features extracted from these silhouette maps are expected to be noisy with distributions of different behavioral classes may overlap. A number of early works have shown the merits of using geometrical model [3]. However, the need to have a high resolution foreground target makes this approach generally less applicable in visual surveillance context. Instead, our problem requires a robust scheme that can address the overlapping of different feature classes for accurate pattern classification.

In the following, we present reviews in the fields of background subtraction and behavior and activity recognition.

1) *Relevant Works on Background Subtraction:* Existing methods of background modeling can be broadly classified into: i) pixel-based methods and ii) block-based methods. One example of pixel-based method is Pfister system by [5], which demonstrated good results to perform real-time tracking by constructing each background pixel as a single Gaussian distribution in YUV color space. When addressing outdoor surveillance (on the ground), recent efforts by [4], [9], and [10] proposed mixture of Gaussians for background modeling. Besides, non-parametric modeling is also found popular among the works by [7] and [19]. One common limitation of these mentioned pixel-based techniques is that they are generally not meant for the purpose to address situation with significant background movements. Thus, these methods are generally more confined to environment with less background movements, e.g., within controlled indoor and outdoor environments.

Examples of block-based method can meanwhile be referred to the works by [20] and [21]. The need of using image block is one limitation of these methods which has constrained the detection to be in block-based unit. As a result, these methods are limited to coarse detection.

In view of aquatic background that exhibits random homogeneous blob movements, we propose in this work a modeling of background based on a set of block-derived homogeneous region units. By matching a pixel of a current frame to models of homogeneous regions within a search window, this can produce good performance of detecting foreground while reducing false detection due to background movement.

2) *Relevant Works on Recognition of Behaviors and Events in Visual Surveillance Context:* Among those works on surveillance and human behavior recognition, it is noted that rule-based method has been a popular approach, owing to its simplicity to establish a knowledge-based framework. A good example of rule-based approach can be referred to the work by [12], which comprises a high-level module that operates on symbolic data and some heuristic rules. One advantage of rule-based approach is that it provides the flexibility of having a reasoning module which could be progressively improved by adding new rules. This provides a simple yet convenient syntax for interfacing with expert knowledge. However, one limitation is its simplicity at each decision level which often relies on simple conditional rules. Thereby, information is not being efficiently utilized.

In contrast to rule-based technique, probabilistic graph models such as Hidden Markov Models (HMMs) have also been receiving enormous attentions for modeling state transition process in temporal domain [22]–[25]. For example, in a recent work by Oliver *et al.* [25], Coupled-HMMs were exploited to model patterns of human activities. However, one common limitation of the mentioned HMM-based methods is that these methods provide little semantic information about behaviors being modeled; instead, semantic information is learnt indirectly through low-level features extracted from training sequences.

In our problem, it is noticed that drowning research has a long history among the lifeguarding community [26], [27]. A drowning incident can be understood as a constellation of nervous response due to suffocation in water, which then trig-

gers external and internal movements known to be *instinctive drowning response*. Therefore, we aim to develop a framework that can classify different behaviors based on the traits of each behavioral class. Visual indicators used by lifeguards for drowning detection are important prior knowledge in our case. In the proposed framework, expert knowledge with semantic meanings are incorporated through a set of defined foreground descriptors. Data fusion and HMMs modeling are applied to learn the state transition of different behaviors, while addressing overlapping of feature classes from different behaviors. Through this framework, we hope to combine the advantages of rule-based approach—that yields a simple and efficient syntax for incorporating semantic meanings, and hidden markov model approach—that yields a probabilistic framework.

C. New Contributions Over Early Works by the Team

This paper shows significant improvements and efforts made compared with our previous works reported in [28] and [29].

- In this paper, we report a transfer of surveillance technology out of lab setting to a real-world problem. To our best knowledge, the proposed DEWS system is the first *live* system using overhead cameras to detect early drowning incidents at pool [14].
- Technical challenges of monitoring and recognizing activities in aquatic environment are identified. A combination of methods that are variations in the fields of background subtraction, denoising, data fusion, state transition and blob splitting are developed.
- Compared with our early works by the same team published in [28], [29], these early works were done on limited short video footages in lab evaluation towards illustrating the idea. But, in this paper, technical challenges under real environment settings are dealt with and new methods are proposed to address long hour operation under different uncontrolled pool environments.
- The foreground detection module in this work was reported earlier in [30], [31]. The focus of these early papers was on object detection in pool environment. Instead, in this paper, the emphasis is on the development of a live surveillance to detect early drowning incidents. In particular, studies on detecting water crisis, treading and numerous swimming activities are reported. Besides, more results and new insights into foreground detection under *crowded scenario* are presented.

III. SCENE MODELING AND FOREGROUND DETECTION IN AQUATIC BACKGROUND

The first processing block of the proposed system comprises: 1) a background modeling module; 2) a foreground detection module; and 3) a specular reflection removal module for detection of foreground at daytime as well as nighttime.

A. Region-Based Background Modeling and Subtraction

Our proposed approach is to model background as a composition of dynamic region-based processes. This is to account for aquatic background, which constitutes homogeneous patches that exhibit random movements in space as well as variations in illumination and color.

Consider a sequence of $N_1 \times N_2$ background frames, and each background frame is divided into $n_1 \times n_2$ non-overlapping square blocks of size $w \times w$ each, i.e., $n_1 = (N_1/w)$ and $n_2 = (N_2/w)$. Denote $\mathbf{X}_{u,v}$ as a set of color vectors collected from square blocks- (u,v) , where $1 \leq u \leq n_1$ and $1 \leq v \leq n_2$. And, $\mathbf{X}_{u,v}$ is to be decomposed into c number of homogeneous regions $\{\mathbf{R}_{u,v}^1, \dots, \mathbf{R}_{u,v}^c\}$. The proposed modeling of background is defined as a composition of the statistical models of $\{\mathbf{R}_{u,v}^k\}$, $\forall u, v, k$.

Background models: Let $\mu_{\mathbf{R}_{u,v}^k} = \{\mu_{\mathbf{R}_{u,v}^k}^1, \dots, \mu_{\mathbf{R}_{u,v}^k}^d\}$ and $\Sigma_{\mathbf{R}_{u,v}^k}$ be the mean and covariance matrices of color vectors of $\mathbf{R}_{u,v}^k$, and $d = 3$ is the dimension of the color space. By modeling color distribution of each $\mathbf{R}_{u,v}^k$ as a single multivariate Gaussian distribution, the probability of a pixel at (i, j) belonging to $\mathbf{R}_{u,v}^k$ can be computed based on similarity in color

$$P(\mathbf{x}_{i,j} | \mu_{\mathbf{R}_{u,v}^k}, \Sigma_{\mathbf{R}_{u,v}^k}) = \frac{1}{(2\pi)^{d/2} |\Sigma_{\mathbf{R}_{u,v}^k}|^{1/2}} \times \exp \left\{ -\frac{1}{2} (\mathbf{x}_{i,j} - \mu_{\mathbf{R}_{u,v}^k})^T \Sigma_{\mathbf{R}_{u,v}^k}^{-1} (\mathbf{x}_{i,j} - \mu_{\mathbf{R}_{u,v}^k}) \right\} \quad (1)$$

where $\mathbf{x}_{i,j} = \{x_{i,j}^1, \dots, x_{i,j}^d\}$ denotes color vectors. Without compromising much the accuracy, we further assume that each color dimension is independent. This gives simplification of (1) to be

$$P(\mathbf{x}_{i,j} | \mu_{\mathbf{R}_{u,v}^k}, \sigma_{\mathbf{R}_{u,v}^k}) = \prod_{m=1}^d \frac{1}{\sqrt{2\pi} \sigma_{\mathbf{R}_{u,v}^k}^m} \exp \left\{ -\frac{(x_{i,j}^m - \mu_{\mathbf{R}_{u,v}^k}^m)^2}{2 (\sigma_{\mathbf{R}_{u,v}^k}^m)^2} \right\} \quad (2)$$

where $\sigma_{\mathbf{R}_{u,v}^k} = \{\sigma_{\mathbf{R}_{u,v}^k}^1, \dots, \sigma_{\mathbf{R}_{u,v}^k}^d\}$ is the standard deviations matrix of $\mathbf{R}_{u,v}^k$.

Therefore, the color similarity between a pixel and a homogeneous region can be computed based on

$$D(\mathbf{x}_{i,j} | \mu_{\mathbf{R}_{u,v}^k}, \sigma_{\mathbf{R}_{u,v}^k}) = \sqrt{\frac{\sum_{m=1}^d \frac{(x_{i,j}^m - \mu_{\mathbf{R}_{u,v}^k}^m)^2}{(\sigma_{\mathbf{R}_{u,v}^k}^m)^2}}{d}}. \quad (3)$$

Through continually updating $\mu_{\mathbf{R}_{u,v}^k}$ and $\sigma_{\mathbf{R}_{u,v}^k}$, a temporal adaptation to background change can be further achieved.

Below are given three implemented steps for the above background modeling scheme.

Step 1—Generating background frames. During initialization, recorded frames are likely to contain both foreground and background. To address the problem, we explore a temporal vector median filter to generate a sequence of background frames.

Let $\mathbf{X}_{i,j}^\tau$ be an array of color vectors collected over τ number of frames, i.e., $\mathbf{X}_{i,j}^\tau = \{\mathbf{x}_{i,j}^t | t = 1, \dots, \tau\}$, where $\mathbf{x}_{i,j}^t$ is the

color vector of t th image at position- (i, j) . A “clean” background frame is generated by applying a vector median filter on $\mathbf{X}_{i,j}^\tau$, $\forall i, j$ that gives

$$\mathbf{B}^1 = \{\mathbf{z}_{i,j} | i = 1, \dots, N_1 \text{ and } j = 1, \dots, N_2\} \quad (4)$$

where $\mathbf{z}_{i,j}$ is computed based on

$$\mathbf{z}_{i,j} = \left\{ \mathbf{x}_{i,j}^q \in \mathbf{X}_{i,j}^\tau \mid \min_{\mathbf{x}_{i,j}^q} \sum_{r=1}^{\tau} |\mathbf{x}_{i,j}^r - \mathbf{x}_{i,j}^q| \right\}. \quad (5)$$

In the process, duration to be taken and efficiency to remove swimmer pixels are two determining factors deciding the sampling rate between two $\mathbf{x}_{i,j}^t$, which has been determined empirically in this work.

To prepare for Step 2 below, background frames $\{\mathbf{B}^l | l = 2, \dots, \kappa\}$ are then generated by applying (5) using a sliding window of $\{\mathbf{x}_{i,j}^t | t = l, \dots, \tau + l\} \forall i, j$.

Step 2—Generating initial background models. After obtaining $\{\mathbf{B}^l | l = 1, \dots, \kappa\}$, the region-based background modeling are implemented by:

- i) Dividing each $\mathbf{B}^l$ into $n_1 \times n_2$ non-overlapping $w \times w$ square blocks, and
- ii) Applying a hierarchical k -means [32] on each $\mathbf{X}_{u,v}$ to obtain cluster centers $\mu_{\mathbf{R}_{u,v}^k}$ and standard deviations $\sigma_{\mathbf{R}_{u,v}^k}$

of homogeneous regions $\mathbf{R}_{u,v}^k$, $\forall k$.

The clustering process starts by initiating one dominant data cluster. In the subsequent iterations, smaller and more compact data clusters are formed through splitting. The process of splitting is terminated when the distance of any two closest cluster centers reaches a predetermined value or the number of cluster centers is greater than 5, whichever is achieved first. Therefore, this forms c -number of homogeneous regions $\mathbf{R}_{u,v}^k$ that gives the initial background models $\mathbf{C}_{u,v}$ to be

$$\mathbf{C}_{u,v} = [\mu_{\mathbf{R}_{u,v}^k} \ \sigma_{\mathbf{R}_{u,v}^k}], \text{ where } k = 1, \dots, c. \quad (6)$$

Typically, the value of c ranges from 2 to 5.

Step 3—Updating background models. To adapt to changes in an outdoor environment, we adopt a recursive linear interpolation scheme for continual updating which shares certain degrees of similarity as the scheme used in [9], [10]. In the updating step, parameters $\sigma_{\mathbf{R}_{u,v}^k}$ and $\mu_{\mathbf{R}_{u,v}^k}$ at time- t is updated according to

$$\mu_{\mathbf{R}_{u,v}^k}^t \leftarrow (1 - \rho) \mu_{\mathbf{R}_{u,v}^k}^{t-1} + \rho \mu_{\mathbf{R}_{u,v}^k}^t \quad (7)$$

$$\sigma_{\mathbf{R}_{u,v}^k}^t \leftarrow (1 - \rho) \sigma_{\mathbf{R}_{u,v}^k}^{t-1} + \rho \sigma_{\mathbf{R}_{u,v}^k}^t \quad (8)$$

where $\rho = 1/\kappa$ is the learning factor for adapting current changes. Parameter κ is defined previously in Step 1, which is the number of “clean” background frames generated. Equivalently, we can interpret (7) and (8) as the averages of all $\mu_{\mathbf{R}_{u,v}^k}^t$ and $\sigma_{\mathbf{R}_{u,v}^k}^t$ of $\mathbf{R}_{u,v}^k$ over a sliding window of length κ . In practice, the updating process is performed by considering only pixels within a threshold distance from the mean. This is to ensure that each Gaussian background model is being adopted slowly while avoiding corruption by outliers. In order to maintain a dynamic background modeling, the ability to create and destroy unnecessary distributions is implemented by assigning a lifespan of κ to each distribution.

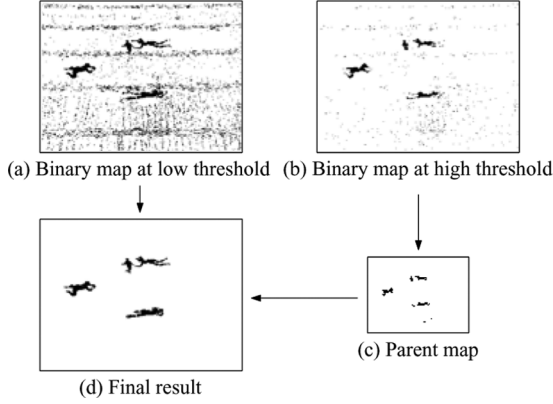


Fig. 4. Thresholding step that combines principles of thresholding-with-hysteresis, denoising and connected component grouping.

B. Foreground Swimmer Detection and Modeling

To account for aquatic background, we explore a foreground detection scheme that incorporates active background movements and an enhanced thresholding. Let $\mathbf{x}_{i,j}$ be the color vector of pixel at location- (i, j) in the square block- (u, v) . Using (3) and considering movements of pixel within a local window, color discrepancy of a pixel and background models is computed to be

$$D_{i,j}^{\min} = \min \left\{ D \left(\mathbf{x}_{i,j} | \mu_{R_{u+q,v+r}^k}, \sigma_{R_{u+q,v+r}^k} \right) \right\} \quad (9)$$

where q and r determine a local window in square-block unit. As observed in experiments, the consideration of neighboring blocks in (9) has contributed to reduction of mis-classification errors at regions, such as areas around lane dividers.

Fig. 4 illustrates the thresholding module we have implemented. The module involves firstly applying a low-threshold value to yield a good segmentation of foreground targets. To suppress noise, a high-threshold map and a parent map are formed. Parent map is a low resolution version of the high-threshold map, where its each element corresponds to a 4×4 non-overlapping block in the high-threshold map. An element of the parent map is labeled as “1” when more than 50% pixels of its corresponding 4×4 block in the high-threshold map are of $D_{i,j}^{\min}$ greater than the high-threshold. The final detection map is lastly formed which comprises connected blobs of low-threshold map that also gives smaller connected blobs in the parent map. This yields an enhanced detection performance with a lower number of false positive as shown in Fig. 4(d).

Updating swimmer models: In the same way as forming the background model, region-based swimmer models are computed and updated every frame once a swimmer is detected. During the online processing phase, color discrepancy measures between pixel’s color and neighboring foreground models, denoted by $D_{i,j}^{f \min}$, are computed. Pixels with $D_{i,j}^{f \min} < D_{i,j}^{\min}$ will be classified as the highest confident swimmer pixels.

C. Swimmer Detection With Filtering Scheme for Nighttime

Operation at nighttime in our problem faces challenges posed by *glare* and *reflection* due to artificial lighting. Polarizer is a well-known technique to suppress virtual layer and reflection in the branch of photographic study [33] by setting camera at

the Brewster angle [34]. However, we face difficulty to achieve good results using the Brewster angle in our system. One reason can be attributed to water surface which is not a plan surface. Additionally, there is difficulty due to inaccessibility of cameras which are mounted high at building structure. Instead, we resort to a simple filtering approach to address the issue. The main idea here is firstly to have a pixel classification step. Subsequently, appropriate filtering actions are applied accordingly for different classes of pixel. Statistical study in temporal domain suggests that the rate of pixel’s color or intensity fluctuation in temporal domain is a useful feature in our problem for the classification step. Denote $I_{i,j}^{t-\xi}, I_{i,j}^{t-\xi+1}, \dots, I_{i,j}^t$ as the intensity values of a pixel at spatial location (i, j) for $\xi + 1$ consecutive frames. A heuristic *fluctuation count* is defined to be

$$\varpi_{i,j}^t = \frac{1}{FR} \sum_{u=0}^{\xi} \Xi_{i,j}^u$$

$$\Xi_{i,j}^u = \begin{cases} 1, & \text{if } |I_{i,j}^t - I_{i,j}^{t-\xi+u}| > \Upsilon \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where FR is the frame rate of video. Condition $|\cdot| > \Upsilon$ provides the measure of significant change in intensity.

With the computation of $\varpi_{i,j}^t$, we classify each pixel into one of the three pre-identified classes using two empirical thresholds η_1 and η_2 , where η_1 and η_2 are empirically set to be one third and two third of ξ/FR in our context. Typically pixels with $\varpi_{i,j}^t > \eta_2$ are fast moving part of foreground. A spatial mean filter with window dimension of 3×3 is employed by only considering pixels within the window with the same ($\varpi > \eta_2$) condition. The mean filtering operation serves the purpose to remove any sensor and digitization noise. On the other hand, there is no filtering if $\varpi_{i,j}^t \leq \eta_1$, which avoids unnecessary blurring on background.

For the class of $\eta_1 < \varpi_{i,j}^t \leq \eta_2$, it contains mostly corrupted foreground pixels due to specular reflection and glare. In this step, we aim to generate a “pseudo color” based on spatio-temporal filtering operation. This is done by firstly computing a set of mean filtered pixels W by applying a 3×3 spatial mean filter centers at (i, j) to the current and a number of successive previous frames. Only pixels with the same ($\eta_1 < \varpi \leq \eta_2$) condition are considered here. Compensated pixel value is then computed to be the median of W . Thus, the overall process provides color estimation based on dominant color of pixels from the class of $\eta_1 < \varpi_{i,j}^t \leq \eta_2$ over current and a number of successive previous frames.

IV. SWIMMING AND WATER CRISES BEHAVIORS MODELING

The step to recognize different water crisis incidents and swimming behaviors involves the use of foreground detection results from Section III. This section describes the details, which comprises: i) a hierarchical representation of swimmer descriptors that incorporates expert knowledge; ii) a data fusion module; and iii) a Hidden Markov Modeling for describing state transition process of behaviors. One technical challenge addressed here is the issue of high correlation between different classes of behaviors due to behavioral variability and inconsistency in the results of foreground silhouette.

A. Swimmer Descriptors Extraction

1) *Drowning Research by Lifeguarding Community*: In the drowning behavioral studies [26], [27], two possible types of water crisis incidents have been identified: distress and drowning. A distress incident involves victim exhibiting involuntary movements such as active struggling or waving. A drowning incident meanwhile can be understood as suffocation in water that triggers external and unlearned movements, which are known to be instinctive drowning response. Both *distress* and *early drowning* victims exhibit the following common visual indicators.

- 1) There will be instinctive response with repetitive arm movements of extending out and pressing down.
- 2) The body will be perpendicular (vertical up) in water with small movements in horizontal and diagonal directions.
- 3) The period of struggling on water surface is on average within the range of 20 to 60 seconds [27].

In this work, both distress and drowning cases are considered under the same event class where alert is to be triggered.

2) *Definition of Swimmer Descriptors*: After studying footages of swimmers demonstrating normal swimming, treading and simulation of actual water crisis incidents (distress and drowning scenarios), we define the following set of descriptors to characterize the unique trait of each behavior.

- a) *Movement range*, f_{MR} . It is noted that activity of normal swimming typically has a larger spatial displacement range over a period of time comparing with the cases of distress, drowning and treading. Hence, this motivates the computation of a movement range parameter, f_{MR} , which is defined to be the maximum displacement of any two centroids recorded over a given period.
- b) *Speed product*, f_{sp} . To describe the pattern of movement, velocity is a natural choice. A target's velocity at an instance can be computed since its centroid positions in current and previous frames have been recorded. Let $\mathbf{v}^t$ and $\mathbf{v}^{t-1}$ be the velocity vectors at the current and previous instances, respectively, and δ be the angle between $\mathbf{v}^t$ and $\mathbf{v}^{t-1}$. We compute a speed product parameter to be $|\mathbf{v}^{t-1}||\mathbf{v}^t|\cos(\delta/2)$, where $|\cdot|$ denotes the magnitude of the velocity vector. The cosine term is to impose penalty to chaotic movement which is observed in the cases of distress and drowning. The division by two in the cosine term confines the angle to be in the range between 0° and 180° .
- c) *Posture variation*, f_{pos} . The angle between the principle axis of computed best-fit-ellipse and the horizontal axis gives the orientation of a swimmer. Since a drowning victim typically shows repetitive arm and leg movements, it corresponds to a continual change in the swimmer's posture. Let $\theta_{\min}$ and $\theta_{\max}$ be the minimum and maximum values of the orientation angles computed over a given period. We hence measure the posture variation to be $(\theta_{\max} - \theta_{\min})$. A high-value f_{pos} is expected when there is a struggling event.
- d) *Activity variation*, f_{act} . Arm and leg movements of extending out and contracting in during distress and early drowning incidents typically lead to a change in numerical value of the ratio between area of best-fit-ellipse and

size of foreground silhouette. We hence compute f_{act} to be the difference between maximum and minimum values of these ratios over a temporal window.

- e) *Size variation*, f_s . We also compute $f_s = (A_{\max} - A_{\min})/A_{\text{mean}}$, where $A_{\max}$, $A_{\min}$ and A_{mean} denote maximum, minimum and average of foreground silhouette's sizes recorded over a temporal window. A high value of this measure is typically observed for distress and drowning cases, due to less consistency in foreground size resulted by water surface disturbance.
- f) *Submersion index*, f_{sub} . This measure is defined to infer the position of a swimmer to be below or above water surface. A sinking swimmer more commonly exhibits a higher color saturation, which adds a blanket of bluishness to skin appearance. The submersion index is computed to be the difference between a swimmer's mean color saturation and minimum saturation value of the swimmer since it is being tracked.

Fig. 5(b)–(g) show plots of the above descriptors in temporal domain when target is in different stages. Fig. 5(h) shows the likelihood plots when classifying the target to be in swimming mode, in treading mode as well as in alert mode. The following two subsections will explain the details to compute these likelihood plots.

B. Data Fusion

Fig. 6 shows scatter plots of defined swimmer descriptors for events of swimming, treading and water crises (distress and drowning scenarios). It shows high correlation among these events where swimmer descriptors of one type overlaps with that of others. To ensure a more accurate decision based on these descriptors, the problem demands more than just a simple linear decision hyper surface. Thus, we explore data fusion technique in this section to yield better information fusion.

Several high performance data fusion techniques are considered for best achievable results. Reduced model (RM) [35] is a relatively new classifier proposed recently by our team in the application of biometrics, which has shown potential due to its fast single step computation without compromising much the accuracy performance. Hence, this motivates us to extend our study of this algorithm in this problem.

Denote $\Phi = [\phi_1, \dots, \phi_6]^T$ be a set of swimmer descriptors in vector form extracted from silhouette of a foreground with label, i.e., $\phi_1 \equiv f_{MR}; \dots; \phi_6 \equiv f_{sub}$. The RM classifier is expressed as

$$g_{RM}(\alpha, \Phi) = \alpha_0 + \sum_{k=1}^r \sum_{j=1}^l \alpha_{k,j} \phi_j^k + \sum_{k=1}^r \alpha_k \left(\sum_{j=1}^l \phi_j \right)^k + \sum_{n=2}^r \left(\sum_{i=1}^l \alpha_{n,i} \phi_i \right) \left(\sum_{j=1}^l \phi_j \right)^{k-1} \quad (11)$$

where ϕ_j , $j = 1, \dots, l$ are the polynomial inputs, $\alpha = [\alpha_0, \alpha_{k,j}, \alpha_k, \alpha_{n,i}]$ are the weighting coefficients to be estimated. Parameters l and r correspond to input-dimension and order of the RM classifier, respectively. As can be seen from (11), the basic component of this polynomial model boils

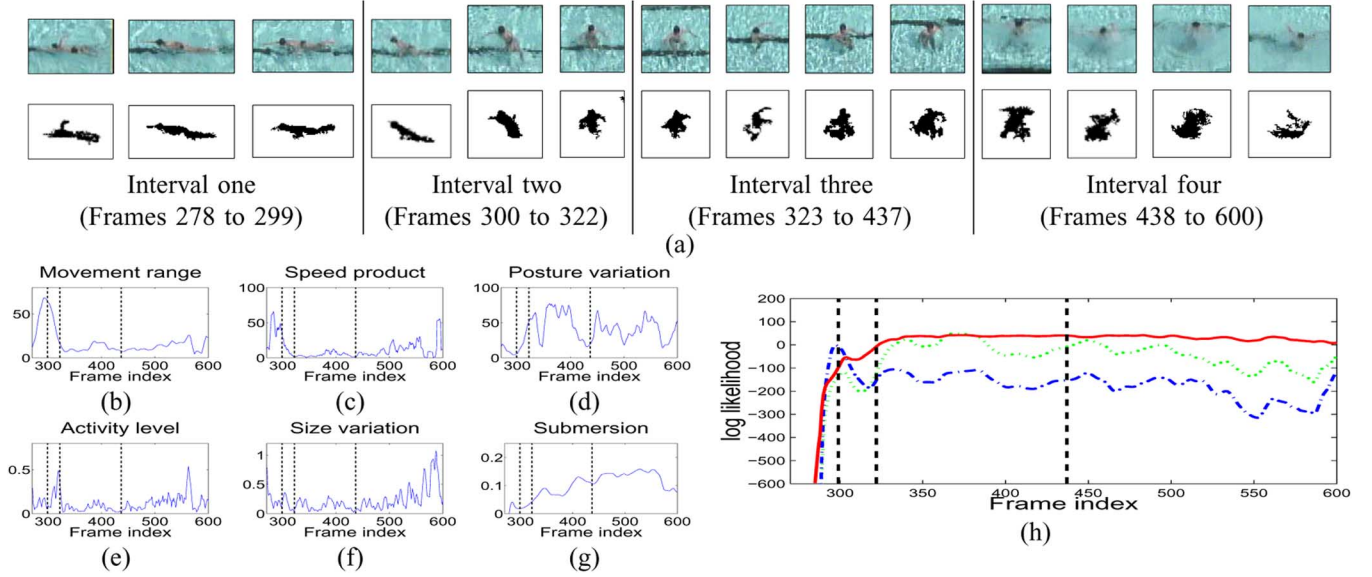


Fig. 5. Simulation of a realistic water crisis incident involves stages: swimming $\rightarrow$ slowing down $\rightarrow$ gentle struggling $\rightarrow$ distress. (b) to (g) show the corresponding plots of the defined descriptors in temporal domain. (h) shows the likelihood plots when classifying the target to be in swimming mode (denoted by dash-dotted line “— · —”), treading mode (denoted by dotted line “...”) and alert mode for possible detection of a water crisis incident (denoted by solid line “—”). The vertical lines divide the plot into the four corresponding intervals given in (a).

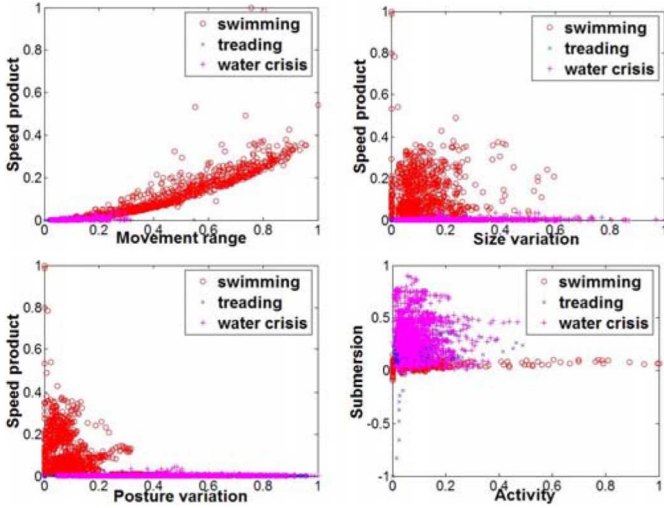


Fig. 6. Scatter plots of defined swimmer descriptors. Data is collected from videos comprising swimmers demonstrate: normal swimming, treading and water crisis (distress and drowning). We consider distress and drowning to be under one class where alert is to be triggered. (Legends: “o” refers to normal swimming, “x” refers to treading and “+” refers to water crisis (distress and drowning)).

down to construction of new pattern features which are sums of the original features, and combination of these new and original features using power and product terms.

By adopting the Least-Squares Error (LSE) objective for computation of α , a simple single step computation has been derived. Given m number of training data points and using a regularized LSE minimization objective

$$R(\alpha) = \sum_{i=1}^m [y_i - g(\alpha, \Phi_i)]^2 + \varrho \|\alpha\|_2^2 \quad (12)$$

the parameter vector α for a 2-class problem (single output target vector $y = [y_1 \ y_2 \ \dots \ y_m]$ containing ‘m’ elements of either “0” or “1” to represent each class) can be estimated from a single step as

$$\alpha = (Q^T Q + \varrho \Lambda)^{-1} Q^T y \quad (13)$$

where $Q \in \mathbb{R}^{m \times K}$ denotes the Jacobian matrix of $g_{RM}(\alpha, \Phi)$, Λ is an identity matrix and ϱ is a small regularization constant.

For a problem with multiple classes as what is being considered, Φ_i is classified to be one of the event classes which is three here. Denote the true events of water crisis (distress and drowning), treading and normal swimming incidents as $y^T \in \{[1 \ 0 \ 0], [0 \ 1 \ 0], [0 \ 0 \ 1]\}$, respectively. Let the target training vectors packed as

$$Y = [y_1 \ y_2 \ \dots \ y_m]^T \quad (14)$$

where each sample row of $Y \in \mathbb{R}^{m \times 3}$ contains only a “1” for the corresponding class and “0” otherwise. Having learned

$$\Theta = (Q^T Q + \varrho \Lambda)^{-1} Q^T Y \quad (15)$$

in a single step, the multi-class model outputs for test can be computed as

$$\hat{Y}_{\text{test}} = Q_{\text{test}} \Theta \quad (16)$$

using Q_{test} generated from test set. For each test data sample (each row of $\hat{Y}_{\text{test}}$), the largest row element will determine the output pattern class.

C. State Transition Modeling

Data fusion step in the previous section provides classification at every instance. Temporal information, which is another important aspect of behavior analysis, has been missing in our consideration so far. To proceed, we formulate the problem to

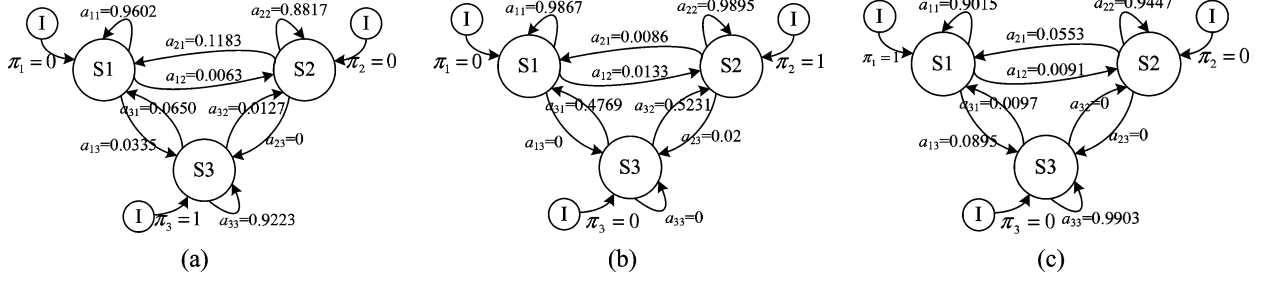


Fig. 7. HMM models describe state transition process of the considered behavior classes. (a) Distress and drowning. (b) Swimming. (c) Treading.

identify the most likely state-transition process of a behavior within a local temporal window.

HMM Modeling: The problem is formulated by considering a three discrete states of HMM [22]–[25], [36], corresponding to the output of the RM classifier. A state variable at time t is given by $s^t \in \{S1, S2, S3\}$. Denote the state transition probabilities as $A = \{a_{u,v}\}$, the observation probabilities as $B = \{b_v\}$, and the initial probabilities being in the respective state as $\pi = \{\pi_u\}$. With the definition of HMM model $\lambda = (A, B, \pi)$, the problem is formulated to identify a state sequence $S = s^1, s^2, \dots, s^T$ from a sequence of observations $O = o^1, o^2, \dots, o^T$ so that $P(O, S|\lambda)$ is maximized

$$S = \operatorname{argmax}_{\{s^t\}_{t=1}^T} P(O, s^1, \dots, s^T | \lambda). \quad (17)$$

Fully connected HMMs are considered in our context. By referring to Fig. 6, we model the distribution of $\{\hat{y}_i\}$ for each analyzed behavior using one single multivariate distribution. Therefore, the observation probabilities $B = \{b_v\}$ could be statistically defined with each b_v as follows:

$$b_v = \frac{1}{(2\pi)^{d/2} |\Sigma_v|^{1/2}} \times \exp \left\{ -\frac{1}{2} (\hat{y}_i - \mu_v)^T \Sigma_v^{-1} (\hat{y}_i - \mu_v) \right\} \quad (18)$$

where Σ_v and μ_v is the covariance and mean of $\{\hat{y}_i\}$ for each behavior class. The transition probabilities are determined through learning training sequences of normal swimming, treading and water crisis (distress and drowning). (See Fig. 7.) Once the HMMs are specified, the state sequence that gives $\operatorname{argmax}\{P(O, S|\lambda)\}$ could be obtained by using Viterbi Algorithm [37].

We have performed a ten-folds validation to decide on parameters used for the above data fusion and HMM modeling schemes. The experiment of this validation process is detailed in Section VI-C.

V. WATER CRISIS DETECTION IN A CROWDED SCENARIO

When addressing a crowded scenario, a higher level of complexity is involved due to *occlusion*. In an occlusion event, silhouette mask of close foreground targets may overlap. This hence results formation of large contiguous blob with foreground targets become inseparable. In order to extend our drowning detection engine to moderate level of crowded scenario, we describe in this section a blob splitting mechanism developed.

In the problem formulation, we view silhouette map of a foreground target as a cluster of labeled foreground pixels where clustering and model-fitting concepts are deployed. To simplify the discussion, the following discusses the splitting step for one detected large contiguous blob. When processing a frame, such process will be repeated for any detection of contiguous blob. Denote a contiguous blob in frame- t as M^t , which is the union of silhouette maps belonging to a group of close foreground targets. Let also γ_i^t be the parameters vector characterizing an ellipse-shape model for foreground- i . The problem involves a process to identify a set of $\Gamma = \{\gamma_i^t\}$ such that this splits a contiguous blob into smaller ellipses. Equivalently, as reported in some relevant earlier works such as [38], this can be formulated to maximize *a posteriori* given by

$$\begin{aligned} \Gamma^{t*} &= \operatorname{argmax}_{\Gamma^t} P(\Gamma^t | \Gamma^{t-1}, M^t), \\ &= \operatorname{argmax}_{\Gamma^t} P(M^t | \Gamma^t) P(\Gamma^t | \Gamma^{t-1}). \end{aligned} \quad (19)$$

To measure the goodness of fitting $\{\gamma_i^t\}$ to M^t , the aim here is to find a solution such that $\{\gamma_i^t\}$ in the image lattice space to give: i) a good coverage that includes as many considered foreground pixels but fewer background pixels and ii) there should be minimum overlapping between any two ellipse-shape models to ensure the visibility of each ellipse. With the stated objectives, we propose a blob splitting mechanism comprising the following steps.

Step 1—Initialization of splitting M^t into k smaller blobs.

- i) The process starts by performing hierarchical k -means clustering on a data matrix, which comprises coordinates of foreground pixels. Let the number of cluster centers be k . Since the clustering step involves minimizing a function of spatial distances between cluster centers and foreground pixels, this yields cluster centers converging to the respective centroid of k -targets. Equivalently, this fits k -number ellipses centered at obtained cluster centers. Since Euclidean distance is used in this initialization step, this gives an initial fitting with round-shape models as presented in Fig. 8(a).
- ii) With the obtained initial cluster centers from (i), foreground pixels are classified and labeled to the respective nearest cluster center. The ellipse-models fitting is then improved by using criteria: Mahalanobis distance of a pixel to its nearest cluster center is less than a threshold. An empirical value of 1.8 has been applied that gives consistent fitting results as shown in Fig. 8(b)–(d).

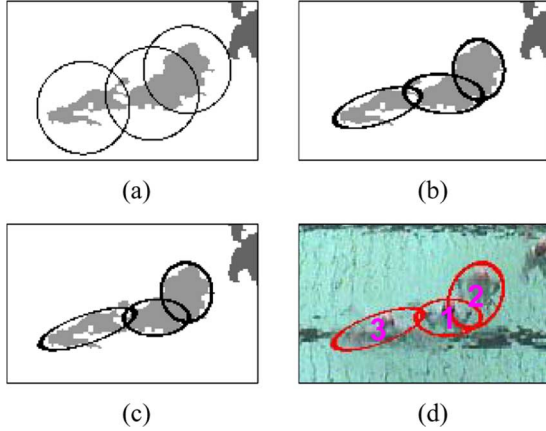


Fig. 8. Steps taken in blob splitting process. (a) Clustering based on Euclidean distance that gives round-shape fitting; (b) Improved fitting by computing Mahalanobis distance and computing ellipse parameters $\{\gamma_i\}$; (c) Parameters of ellipses $\{\gamma_i\}$ are perturbed to give better fitting; (d) Final result of the blob splitting process.

Step 2—Selection of the best combination of ellipses. This step involves a recursive process where parameters of ellipses $\{\gamma_i\}$, $i = 1, \dots, k$ are perturbed to provide different likely hypotheses. A validity score defined based on the likelihood and prior components of (19) is then computed for each hypothesis to allow selection of the best solution.

- i) The perturbation starts by considering an ellipse- j , $j \in i$. Pixels in the overlapping area between ellipse- j and other ellipses are re-assigned based on nearest distance criteria of these pixels to other cluster centers. This allows ellipse- j to shrink while other ellipses to expand, generating new solution to the fitting. Likelihood and prior measures are then computed using (20) and (21) defined below

$$P(\mathbf{M}^t | \Gamma^t) \propto \sum_{i=1}^k \frac{A_{\gamma_i}^{bg} + A_{\gamma_i}^{ov}}{A_{\gamma_i}^{fg}} \quad (20)$$

where $A_{\gamma_i}^{bg}$ and $A_{\gamma_i}^{fg}$ give background and foreground pixel counts within the area defined by γ_i^t , and $A_{\gamma_i}^{ov}$ gives foreground pixel count within the overlapping area of ellipse- i and other ellipses. And

$$P(\Gamma^t | \Gamma^{t-1}, \mathbf{M}^t) \propto \varphi |k^t - k^{t-1}| \quad (21)$$

where k^t denotes the current number of targets and k^{t-1} denotes the number of targets formed in the previous frame associated to $\mathbf{M}^t$ in the current frame. Parameter φ serves as an empirical weight. Therefore, a validity score is computed as the product of (20) and (21).

- ii) The above Step (i) is then repeated by selecting another value for j and the corresponding validity score is recorded. An intermediate solution of $\{\gamma_i\}$ is obtained which gives to a minimum value for these computed validity scores.

The above (i) and (ii) of Step 2 are then repeated to continuously improve the fitting. Since the process is aimed to achieve a solution that gives a minimum value of the validity score, the process is terminated when the next validity score is greater than the current validity score. This produces the end results as given in (c) and (d) of Fig. 8.

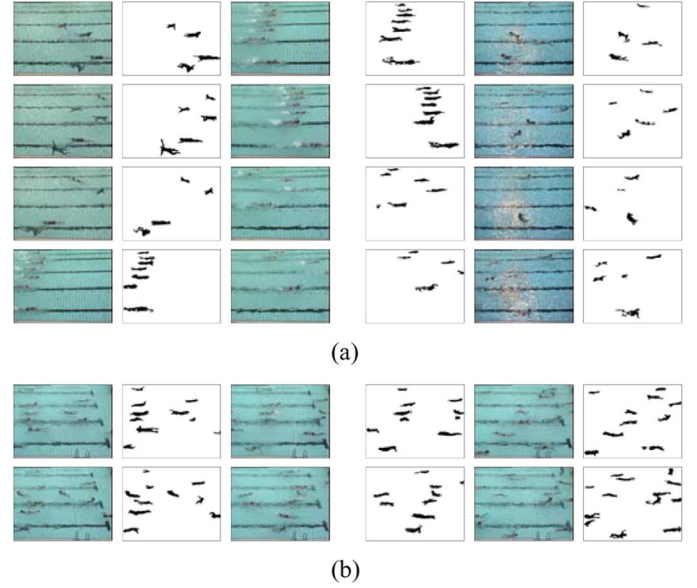


Fig. 9. (Top to bottom, left to right) Detection of swimmers for frames: (a) recorded by a camera at a deeper zone from 8 am to 9 pm, demonstrating different activities in the pool; (b) recorded by a camera at a shallower zone. (Odd columns: Frames recorded at different time intervals; Even columns: Silhouette maps of segmented foreground). (a) The deeper zone (b) The shallower zone.

Step 3: Selection of best k . In order to select the best k , the above Steps 1 and 2 are repeated for a range of k values, where corresponding validity score for each k is recorded. The best k value is decided to be the one which gives the minimum validity score.

VI. EXPERIMENTAL RESULTS

Below are summarized different aspects of analysis performed in the experiment, which comprise the following:

- qualitative and quantitative evaluations of foreground detection on a whole day sequence given in Section VI-A;
- evaluation of foreground detection in challenging situations and its comparison to state-of-the-art background subtraction methods given in Section VI-B;
- accuracy of water crisis and swimming activities detection given in Section VI-C;
- accuracy of water crisis detection in moderate level of crowded situation given in Section VI-D;
- overall system performance of an on-site trial given in Section VI-E.

In the system setup, eight overhead cameras have been installed to divide the pool equally into eight zones. Since cross cameras tracking is not a focus of this work, we have ensured that there is overlapping between the views of any two adjacent cameras. This is to account for the case where an event of interest may happen on the boundary between two cameras' views. The proposed system runs at 4 frames/s under different ambient variations and at different levels of crowd density and activity.

A. Foreground Detection on a Whole Day Video

1) *Qualitative Evaluation:* Fig. 9 presents results of continuous running on a whole day sequence from 8 am to 9 pm.

TABLE I
QUANTITATIVE EVALUATION ON VIDEO SEQUENCES IN FIG. 9. IN THIS EVALUATION, MANUAL INSPECTION WAS PERFORMED TO DERIVE THE GROUND TRUTH. (LEGENDS: TP REPRESENTS TRUE POSITIVE; FN REPRESENTS FALSE NEGATIVE; FP REPRESENTS FALSE POSITIVE)

		TP (%)	FN (%)	FP (%)
Deep Zone	Morning	≈ 98	≈ 2	≈ 0
	Daytime	≈ 99	≈ 1	≈ 1
	Nighttime	≈ 95	≈ 5	≈ 10
Shallow Zone	Daytime	≈ 99	≈ 1	≈ 1

As our system operates at 4 frames/s, the figure shows snapshots of different interesting events from the processed 129 600 ($= 9 \times 3600 \times 4$) frames. In (a), images arranged from top to bottom, left to right, depict: 1) a less crowded scene in morning; 2) a high activity scene during a training lesson on afternoon; and 3) a scene with specular reflection at night. In (b), it meanwhile shows images captured by a side camera at a shallower zone. It is noted that visibility of swimmers in these presented images is generally weak attributed by noisy background.

In the figure, obtained foreground silhouette maps are presented side by side comparing with the corresponding original frames. From these results, most foreground targets have been correctly segmented, and consistent performance has been observed over different periods of time.

2) *Quantitative Evaluation*: In Table I, we meanwhile report quantitative results in terms of true positive, false negative and false positive rates for four video sequences, corresponding to the different time intervals and cameras shown in Fig. 9. Each of these sequences is selected to be of length 2000 frames. The ground truth was obtained by manual checking of these 8000 frames.

- 1) The study of morning sequence shows a detection rate of $\approx 98\%$. The $\approx 2\%$ miss detection has been mainly due to low contrast between swimmers and background. In addition, there are swimmers of smaller size which has made the detection a difficult task. A low false positive rate (i.e., $\approx 0\%$) shows the robustness of the proposed algorithm to address dynamic aquatic background.
- 2) The afternoon sequence comprises frames captured during a swimming lesson. A more dynamic background is involved here. In this study, water splashes caused by greater swimming movement is one main factor contributing to the slightly higher false positive rate (wrongly detecting movement of some background patches as foreground).
- 3) We observe deterioration of the performance at nighttime due to strong specular reflection which has led to a noisier environment. The higher false positive rate observed has been mostly attributed by cases of wrongly detecting background as foreground in reflective region.
- 4) The fourth video sequence shows a study at a shallower zone, corresponding to scenario given in Fig. 9(b). Equally good performance of foreground detection has been obtained. This shows the developed algorithm can directly be applied at different zones of the pool.

It is noted that similar analysis as discussed above had been published in our earlier work in [30] and [31]; but on sequences recorded on a different day. The above has shown a new set of

results. Hence, this section serves two purposes: 1) to further reaffirm the performance of our algorithm and 2) to ensure that this paper is self-contained, covering results of low-level processing and the later section on event detection module.

B. Challenging Scenarios and Performance Comparison

We will highlight in this section the performance yielded corresponding to a few challenging scenarios faced and its performance comparison against a few state-of-the-art methods for performance benchmarking.

1) *Case 1: Nighttime Sequence With Specular Reflection*: Fig. 10(a) presents challenge due to specular reflection at nighttime scene. The third row of the figure shows results obtained after incorporating the developed filtering scheme. Improved performance with greater noise immunity and higher detection accuracy is observed. We would like to highlight that there are constraints to use polarizer in an efficient manner in our system setup. Thus, the developed image filtering technique by exploiting momentary characteristic of pixel signal in temporal domain shows an attractive alternative approach.

2) *Case 2: Sequence With Swimmers Playing, Generating Background Disturbance*: Fig. 10(b) depicts a scenario with significant water disturbance. As shown, lane dividers disappear after swimmers jumping into the pool. Such water movement will likely be misinterpreted as foreground when applying conventional approaches. The devised spatial matching mechanism has played a key role here to efficiently recognize such water movement. Promising results have also been observed after the lane dividers reappear again given in the third and fourth rows.

3) *Case 3: Performance Comparison*: Fig. 10(c) shows performance comparison to state-of-the-art methods. When implementing W4 [7] and mixture of Gaussian (MoG) [9], [10] methods, we have carefully tuned these methods to obtain the best results in our context. These results show that our algorithm has yielded a higher efficiency in detecting small and poorly visible swimmer while maintaining a low false positive error. It is understandable that this may not be a fair comparison. However, it reveals that the current state-of-the-art algorithms may not be straightforwardly extended to such new problem, and naturally there is a need for further research.

4) *Case 4: Comments on Other Scenarios*: We observed that images obtained from rainy sequence have a calmer background. This has made the foreground detection task less difficult. In addition, from the operational aspect, accurate foreground detection is less required as swimming is generally not permitted during rainy period.

Due to constraints such as permission of accessing a pool and costs needed, in this paper, we have yet to set up and evaluate the proposed DEWS system at other pool with unusual geometry dimension. However, based on results from the above experiments, we believe that the proposed system can be extended to such a pool with proper number of cameras to avoid any blind spots and to ensure good monitoring views.

C. Off-Line Evaluation of Water Crisis Detection

To perform the experiment, 20 sequences comprising events of *swimming*, *treading* and simulations of actual *water crisis* (distress and drowning) are collected. These sequences have

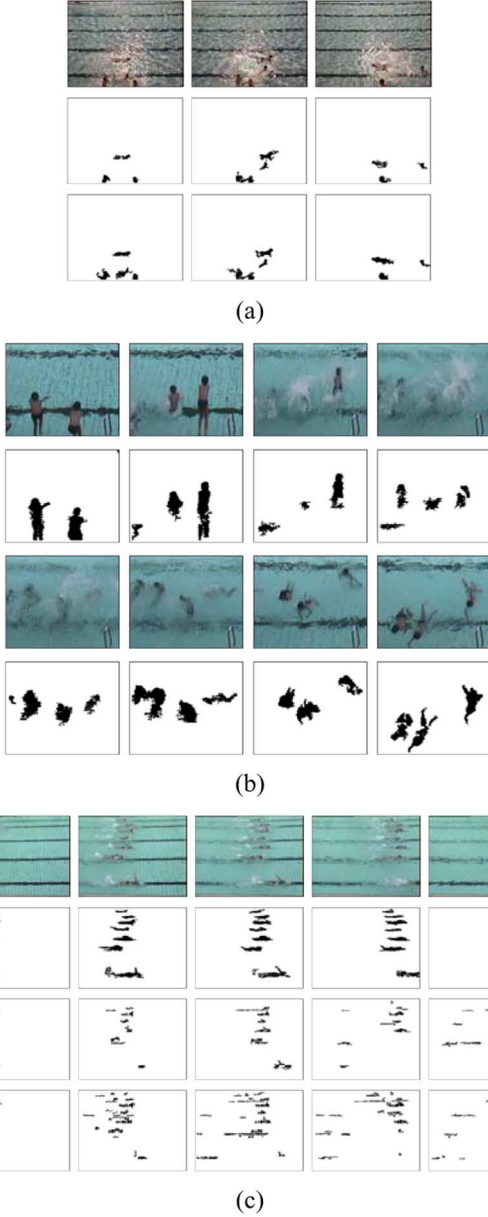


Fig. 10. Results demonstrate: (a) Foreground detection on nighttime scenes for cases without (2nd row) and with (3rd row) the proposed spatio-temporal filtering scheme. (b) A scenario with swimmers playing causes significant background disturbances. (c) Performance comparison: our algorithm in 2nd row, W4 [7] in 3rd row and mixture of Gaussian [9], [10] in 4th row.

been collected from different cameras with each sequence may contain a series of activities as given in Fig. 5. We have manually segmented these sequences to collect approximately 2000, 6000, and 6000 sets of swimmer descriptors for each respective class. This is equivalent to a continuous sequence of approximately an hour in our context.

In this experiment, a ten-folds validation study is carried out: i.e., 90% of each class is selected as training data set while the remaining as testing data set. This process is repeated 10 times with different combinations of training and testing data sets. In the experiment, the ground truth is manually annotated by assigning the three events {water crisis, treading, normal swimming} to be

TABLE II
AVERAGE ERROR RATES OF THE RESPECTIVE CLASSIFIERS BASED ON A 10-FOLDS VALIDATION PROCESS. FOR FNN, SVM AND RM METHODS, TWO SETS OF RESULTS (RUN-1 AND RUN-2) FROM PARAMETER SETTINGS THAT GIVE BEST TRAINING AND BEST TESTING ERRORS ARE PRESENTED. OWM IS A SIMPLE METHOD WHERE A SET OF OPTIMUM WEIGHTS ARE UNIQUELY COMPUTED

	%Error _{train}	%Error _{test}
OWM	≈ 17.59	≈ 18.78
FNN-run-1 (Best training)	≈ 17.64	≈ 21.53
FNN-run-2 (Best testing)	≈ 19.08	≈ 19.08
SVM-run-1 (Best training)	≈ 7.18	≈ 21.71
SVM-run-2 (Best testing)	≈ 11.87	≈ 15.15
RM-run-1 (Best training)	≈ 12.64	≈ 15.70
RM-run-2 (Best testing)	≈ 12.67	≈ 15.57
RM+HMM($T = 16$)	≈ 8.88	≈ 12.28

{[1 0 0], [0 1 0], [0 0 1]}, respectively. For SVM method [42], RBF kernel is used, where we have explored different combinations of gamma values in the range of {0.1, ..., 0.9, 1, ..., 9, 10, ..., 400} and parameter C values in the range of {0.01, 0.1, 1, 10, 100}.

Table II shows average error rates of the ten-folds validation performed. Two sets of results (run-1 and run-2) which give best training and best testing errors are listed in the table. Since OWM [39] is a simple method where a set of optimum weights can be uniquely computed during training, there is only a set of average error rate given in the table.

Table II shows better performance attained by the developed RM with smaller error rates for both the training and testing data sets, comparing with *optimal weighting method* (OWM) [39] and *feedforward neural network* (FNN) [40], [41]. The reason FNN has a larger classification error is due to its convergence to local error solution. The developed RM has also achieved a comparable performance as what achieved by the well-known *support vector machine* (SVM) [42]. One technical merit of developed RM is its fast single computation step while without compromising much the performance accuracy. The incorporation of HMM module which exploits temporal information has further contributed to additional performance improvement. A reduction of about 4% in the error rates is recorded as tabulated.

From the training result that has given the minimum testing error, the order of the RM classifier in (11) and regularization constant in (13) are obtained to be $r = 3$ and $\varrho = 1e - 2$. Meanwhile, HMM models that describe state transition of the analyzed activities are given in Fig. 7.

Figs. 11 and 12 show analysis in time domain. The first two scenarios in Fig. 11(a) and (b) give simulations of *distress* event, where the victims demonstrate attributes having large arm movements to keep body afloat and small lateral movements with body perpendicular to water surface. The details of these experiments are further tabulated in Table III. Parameters L_{wc} , L_{tread} and L_{swim} denote likelihood scores for water crisis, treading or swimming events. The experiment in Fig. 11(a) starts by having treading activity for frames 320–326 and follows by a realistic distress simulation for frames 327–400. The result shows a larger L_{wc} ($L_{wc} > L_{tread}$ and $L_{wc} > L_{swim}$) obtained for frames 340–400, in which an alert has been successfully triggered.

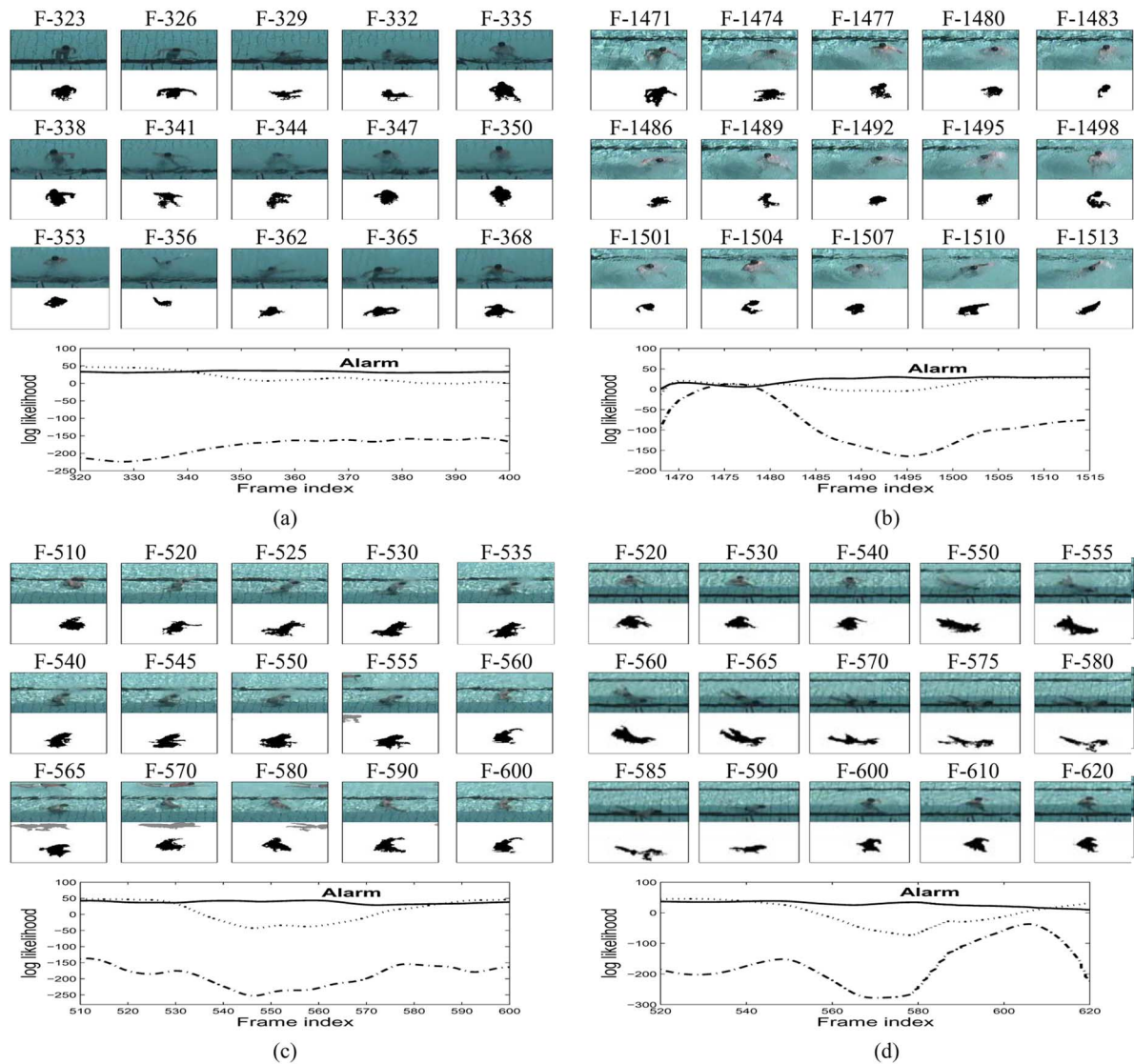


Fig. 11. Detection of simulations of realistic water crisis events (distress and drowning). For simulations of distress in (a) and (b), the victim demonstrates large arm movements but small lateral body movements. Meanwhile, for simulations of drowning in (c) and (d), the victim sinks beneath the water surface without much arm movement. Result shows higher water crisis likelihood yielded when the target is exhibiting distress for (a) and (b) and drowning for (c) and (d). (Legends: solid line "—" denotes water crisis likelihood, dotted line "..." denotes treading likelihood, and dash-dotted line "- · -" denotes swimming likelihood). (a) and (b) Distress. (c) and (d) Unconscious drowning.

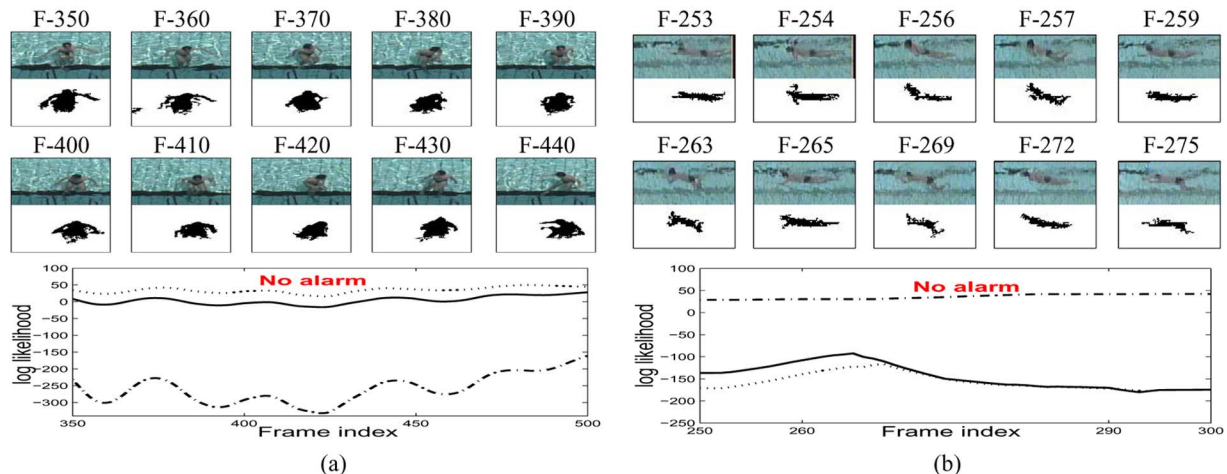


Fig. 12. Detection of treading and swimming activities. (a) Result shows correct detection of the treading activity, demonstrating a higher treading likelihood curve. Knowing the margin between treading likelihood and water crisis likelihood can allow an incorporation of a bias for better water crisis detection while balancing the false alarm. (b) Result shows that swimming likelihood curve is far above treading and distress likelihood curves. (Legends: solid line "—" denotes distress likelihood, dotted line "..." denotes treading likelihood and dash-dotted line "- · -" denotes swimming likelihood). (a) Treading. (b) Swimming.

TABLE III

PRESENTS THE DIFFERENT STAGES OF ACTIVITIES FOR VIDEO SEGMENTS GIVEN IN FIG. 11, AND THE DIFFERENT EVENT CLASSES { NORMAL SWIMMING, TREADING, WATER CRISIS } WHICH HAVE THE HIGHEST LIKELIHOOD SCORES AT DIFFERENT INTERVALS. TIME TAKEN TO TRIGGER ALARM FROM THE ONSET OF WATER CRISIS IS GIVEN IN THE LAST COLUMN. (LEGENDS: L_{wc} AND L_{tread} DENOTE LIKELIHOOD SCORES FOR WATER CRISIS AND TREADING CLASSES, RESPECTIVELY)

	Activities at different intervals	Largest likelihood	Time taken (Alarm triggered)
Fig 11 (a)	F-320 → F-326 (Treading) F-327 → F-400 (Distress)	F-320 → F-340 (L_{tread} is the largest) F-341 → F-400 (L_{wc} is the largest)	9.25s
Fig 11 (b)	F-1468 → F-1473 (Treading) F-1475 → F-1515 (Distress)	F-1470 → F-1480 (L_{tread} is the largest) F-1481 → F-1515 (L_{wc} is the largest)	7.5s
Fig 11 (c)	F-510 → F-519 (Treading) F-520 → F-570 (Drowning) F-571 → F-600 (Treading)	F-510 → F-539 (L_{tread} is the largest) F-530 → F-585 (L_{wc} is the largest) F-586 → F-600 (L_{tread} is the largest)	8.5s
Fig 11 (d)	F-520 → F-537 (Treading) F-538 → F-594 (Drowning) F-595 → F-620 (Treading)	F-520 → F-540 (L_{tread} is the largest) F-541 → F-610 (L_{wc} is the largest) F-611 → F-620 (L_{tread} is the largest)	6.75s

The alarm triggering step involves checking a cumulative score, which is increased by “1” when L_{wc} is the largest compared with L_{tread} and L_{swim} ; otherwise, decreased by “1” until reaching “0.” A swimmer will be labeled the stage of “yellow,” if the cumulative score is more than a value equivalent to three seconds. If such observation is maintained with the cumulative score exceeds a value equivalent to six seconds, the status of the victim will be changed to “red.” Thus, an alert will be automatically triggered. After factoring in the lag due to the temporal window used in the HMM step, time taken to trigger the alarm for this experiment is approximately 9.25 seconds.

Likewise for the case study given in Fig. 11(b), a larger L_{wc} is observed for frames 1481–1515, in which an alarm has been triggered. It is noted that L_{tread} value is close to L_{wc} for frame 1500 onwards. One reason contributed to this observation is less accurate foreground silhouette being generated.

In Fig. 11(c) and (d), we meanwhile show two simulations of *drowning* event. More specifically, these are simulations of unconscious drowning. The main differences of these two simulations from the previous described distress simulations are that there is less arm movement and the event happens with the victim sinks to the bottom of pool with little struggling. In these two case studies, the victims demonstrate treading in the beginning, and then followed by sinking to the bottom of pool. The figure shows good recognition of these swimmers’ activities which matches well to the actual actions. It reports a higher treading likelihood in the beginning of the experiment, and followed by a higher distress likelihood observed in between frames 530–585 for (c) and frames 541–610 for (d). A higher treading likelihood at the end is observed for both cases is due to the simulations are completed with a treading activity. Meantime, swimming action to move to water surface has led to a rise in L_{swim} towards the end of the sequence.

Among activities in pool, it is identified that *treading* activity shares similarity as distress. Fig. 12(a) presents a study of treading event, showing a higher treading likelihood curve. The observed margin between the treading and distress likelihood curves is useful to incorporate an additional bias to enhance the detection of distress while balancing between false detection in the practical implementation.

Fig. 12(b) demonstrates a study of normal swimming event. The figure shows swimming likelihood curve is far above

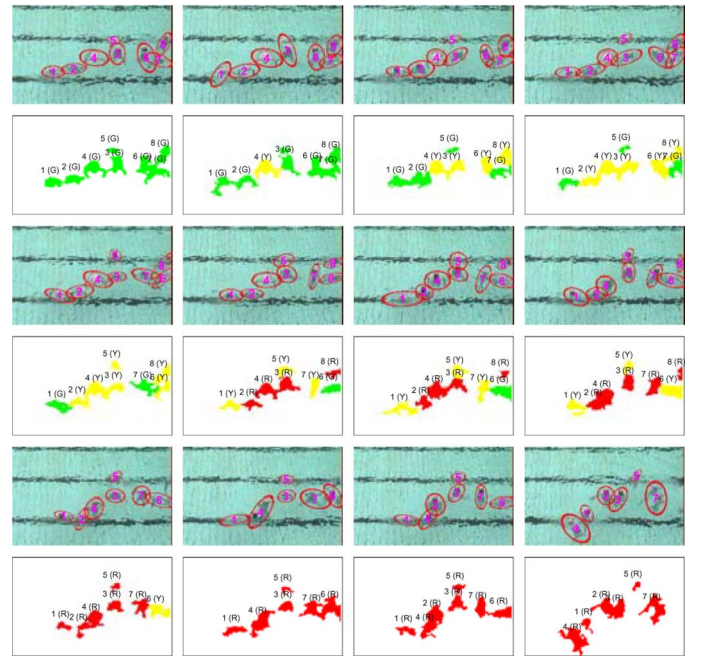


Fig. 13. Sequence images from a life-saving lesson. In the lesson, swimmers demonstrate distress in water. Occlusion happened causes silhouette masks of nearby swimmers overlap, forming a contiguous blob. The above results show the implemented blob splitting mechanism can do reasonably well in separating a contiguous blob into the respective swimmers.

treading and distress curves. This infers that it is less likely of wrongly detecting a normal swimming activity as distress.

D. On-Site Testing: Water Crisis Detection in a Crowded Scenario

Fig. 13 reports results obtained when evaluating the trained system in a life-saving lesson. This is a challenging scenario as it involves a close group of swimmers demonstrating realistic water crisis scenario in water. Occlusion happened has caused formation of large silhouette blob, leading to issues like change in target’s label and destroy of the past cumulated record.

The odd rows of Fig. 13 present sample frames from the sequence. Red circles overlaid on these frames show fitting a k number of ellipses by the developed blob splitting mechanism. The figure shows reasonably good separation obtained where

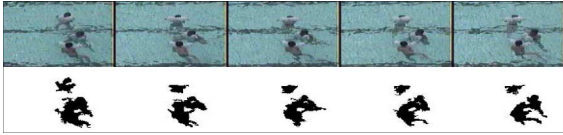


Fig. 14. Detection of a scenario which looks similar to a distress case during the on-site trial.

contiguous blobs have been segregated into individual entities. Since tracking is not a key focus of this work, we have adopted a linear tracking scheme where centroids of detected swimmers from previous frame are linearly projected onto current frame, and the association is done by pairing up centroids in current frame to the projected centroids from previous frame based on shortest distance. In this study, the blob splitting mechanism has contributed to the improvement of cumulating evidence for the event detection step.

In the even rows, individual target's label and its status (i.e., in either "green", "yellow", or "red") are presented. Statuses of all swimmers are in "green" in the beginning, showing no water crisis event. As the struggling action progresses, statuses of swimmer change from "yellow" to "red" eventually, as shown.

E. On-Site Trial: False Alarm Cases

In the current implementation, we ensure the developed system should response to cases which is similar to behaviors/scenarios as have been collected in the training phase. Meantime, the false alarm rate is at about 1 to 5 cases for each camera in a day. We would like to note that these results are subject to activities in the pool and the sensitivity of detecting genuine cases as required by end-users. Fig. 14 shows an example that have triggered the alarm, which involves a group of new learners practising their swimming skill. Based on feedbacks from end-users (lifeguards), triggering alarm for such scenario is desirable as which will draw their attention to higher risk area.

VII. CONCLUSIONS AND FUTURE WORKS

We have presented in this paper a vision system for early drowning detection at an outdoor swimming pool. Based on identified technical challenges, a set of methods which are new variations in the fields of background subtraction, denoising, data fusion and blob splitting technique are proposed. In particular, we have addressed fundamental problems such as : i) the issue of devising a good modeling scheme for dynamic aquatic background; ii) the issue of recognizing behaviors and activities in swimming pool context; and iii) the issue of addressing moderate level of crowded scenario. Promising results yielded in the experiments demonstrate that our proposed DEWS system is beyond the stage of proof-of-concept. Currently, we are working towards delivering a set of operation specifications with inputs from the end-user (lifeguard).

It is also realized that a number of issues remain unexplored in this work, and thus will be considered in our future works. One issue is a drowning incident may happen in a form which can be different from the learned instinctive drowning response model. Therefore, this poses the challenge on how our system will react to an event, which the system is not trained to. On

the other hand, it is also desirable to recognize group activity to complement the current approach which is primarily relying on analysis of activity for every single target.

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